Module 8

JavaScript JavaScript Introduction

Que 1. What is JavaScript? Explain the role of JavaScript in web development.

Ans:- JavaScript is a high-level, interpreted programming language used to create interactive and dynamic web pages. It runs directly in the browser and is one of the core technologies of web development alongside HTML and CSS.

- HTML → Structure (skeleton)
- CSS → Styling (design)
- JavaScript → Behavior (interactivity)

Role of JavaScript in Web Development

1. Adds Interactivity

- Handles user actions like clicks, typing, scrolling
- Example: Button click → show message

2. DOM Manipulation

- DOM = Document Object Model (structure of webpage)
- JavaScript can **change content, styles, and elements dynamically**

3. Form Validation

- Checks user input before submitting forms
- Example:
  - Email format check
  - Required fields validation

4. Dynamic Content Loading

- Updates content without refreshing the page
- Used in modern apps like dashboards, social media

5. Event Handling

- Responds to user actions (events)
  - click
  - hover
  - keypress

Que 2. How is JavaScript different from other programming languages like Python orJava?

Ans:-

1. **Execution Environment**

   JavaScript

   - Runs directly inside the browser (Chrome, Edge, etc.)
   - No installation needed for basic use
   - Can also run on the server using Node.js

   Python

   - Runs on a Python interpreter installed on your system
   - Mainly used for backend, scripting, and data processing
   - Not used directly inside browsers

   Java

   - Runs on JVM (Java Virtual Machine)
   - First compiled into bytecode, then executed
   - Platform-independent ("Write Once, Run Anywhere")

2. **Compilation VS Interpretation**

   JavaScript

   - Interpreted language (modern engines use JIT – Just-In-Time compilation)
   - Code runs immediately in browser

   Python

   - Fully interpreted
   - Slower compared to compiled languages

   Java

   - Compiled + interpreted
   - First compiled → then executed by JVM
   - More efficient for large systems

Que 3. Discuss the use of <script> tag in HTML. How can you link an external JavaScript file to an HTML document?

Ans:- The <script> tag in HTML is used to **add JavaScript code** to a webpage. It allows you to make your website **interactive, dynamic, and functional**.

Why <script> Tag is Used

- Add interactivity (click events, animations)
- Validate forms
- Manipulate HTML elements (DOM)
- Load dynamic content
- Communicate with servers (APIs)

Ways to Use <script> Tag

1. Internal JavaScript (Inside HTML)

You can write JavaScript directly inside the <script> tag.

Example :-

```
<!DOCTYPE html>
<html>
<head>
   <title>Internal JS</title>
</head>
<body>

<h2 id="demo">Hello</h2>

<script>
   document.getElementById("demo").innerHTML = "Hello JavaScript!";
</script>

</body>
</html>
```

2. External JavaScript

You can link a separate .js file using the src attribute.

Example:-

```
<!DOCTYPE html>
<html>
<head>
<title>External JS</title>
```

```html
</head>
<body>

<h2 id="demo">Hello</h2>

<!-- Linking JS file -->
<script src="script.js"></script>

</body>
</html>
```

Variables and Data Types Theory Assignment

Que 1. What are variables in JavaScript? How do you declare a variable using var, let,and const?

Ans:- A **variable** in JavaScript is a **container used to store data values**.
It allows you to save information (like numbers, text, or objects) and reuse or modify it later in your program.

Example Of a Variable

```javascript
let name = "Krunal";
let age = 21;
```

How to Declare Variables in JavaScript

JavaScript provides **three keywords** to declare variables:

var, let, and const

1. Var

```javascript
var x = 10;
```

Features:

- Function-scoped
- Can be **redeclared**
- Can be **updated**

```javascript
var x = 10;
var x = 20; // ✅ allowed
x = 30;     // ✅ allowed
```

2. **Let (Modern & Recommended)**

let y = 15;

Features:

- Block-scoped (works inside {} only)
- Can be **updated**
- Cannot be redeclared in same scope

let y = 10;
y = 20;   // ✅ allowed
// let y = 30; ❌ error

3. **Const (Constant Value)**

const z = 50;

Features:

- Block-scoped
- Cannot be **updated**
- Cannot be **redeclared**

const z = 10;
// z = 20; ❌ error

Que 2. Explain the different data types in JavaScript. Provide examples for each.

Ans:- In JavaScript, **data types** define the kind of value a variable can hold.
They are mainly divided into **two categories**:

1. **Primitive Data Types**

These are **basic, single values** (immutable).

I. **Number**

Represents **numeric values** (integers & decimals)

let age = 25;
let price = 99.99;

II. String

Represents **text data**

```
let name = "Krunal";
let message = 'Hello World';
```

III. Boolean

Represents **true or false**

```
let isLoggedIn = true;
let isAdmin = false;
```

IV. Undefined

A variable declared but **not assigned a value**

```
let x;
console.log(x); // undefined
```

V. Null

Represents **intentional empty value**

```
let data = null;
```

👉 Difference:

- undefined → not assigned
- null → intentionally empty

Ques 3. What is the difference between undefined and null in JavaScript?

Ans:-

undefined	**Null**
Indicates a variable that has been declared but not yet assigned a value.	Represents the intentional absence of any object value.
Primitive data type in JavaScript.	Primitive data type in JavaScript.
Automatically assigned to variables that are declared but not initialized.	Must be explicitly assigned to a variable.
undefined == null // true (loose equality considers them equal).	undefined == null // true (loose equality considers them equal).
undefined === null // false (they are not strictly equal in type).	null === undefined // false (they are not strictly equal in type).
Used by JavaScript when a variable is not assigned a value.	Used intentionally by developers to indicate "no value" or "empty".
undefined is a global property with the value undefined .	null is a global property with the value null .
undefined values are omitted during serialization.	null values are preserved during JSON serialization (e.g., {"key": null}).